# Ten Layers of the Actual First-Alias Replacement Frontier: Synchronized Signed Compensation and an Inactive Determinant Bridge

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**Abstract**

We audit RH-332–RH-340 on one physical natural clock, one Hardy normalization, and the common determinant cut $u = 4k$. The deterministic frontier left open by RH-241 has since split cleanly: RH-263 proves the all-order deterministic numerator anchor, RH-267 proves a certified all-order geometric envelope, and RH-268 proves its sharp coefficient radius. These results do not supply the moving noisy all-order coefficient bridge or Gate A.

For the corrected physical observation type, the source chain is

$$q_n = \mathcal{B}_n + \mathcal{S}_n + \mathcal{R}_n + \mathcal{P}_n - \mathcal{A}_{k,n}, \qquad p_n = q_n - d_n, \qquad |P_u - E_u| \le D_u.$$

At $u = 4k$, both analytic tails vanish, while direct-prefix closure still requires the same-clock conjunction $D_{4k} \to 0$, $E_{\mathrm{off},4k} \to 0$, and $q_{\sigma,k,2k} = o(H_k)$. The physical boundary-orbit atoms at orders $2k$ and $2k-2$ force two exponentially precise signed compensation equations. Separate absolute splitting of orbit, diffuse, and head pieces has a divergent two-atom submajorant.

The new synthesis theorem is strictly about the available information class. Its unconstrained signed combined complements admit an abstract cancelling ledger completion and an abstract noncancelling ledger completion. Thus the current exact identities and asymptotic atom sizes determine neither aggregate prefix closure nor nonclosure. These ledgers are not two physical noisy operators, and no realizability claim is made. The physical prefix, off-alias background, head budget, determinant gluing, and Gates A–E remain open. No Riemann-hypothesis conclusion follows.

## 1 The inherited frontier and the common coordinate

RH-241 ended with two distinct open deterministic-side interfaces: a uniform all-order target envelope and coefficient identification [11]. The later deterministic papers discharge those target obligations. RH-263 gives the all-order deterministic numerator anchor $a_n$ [4]; RH-267 proves

$$|a_n| < 48 q_*^n \quad (n \ge 2), \qquad q_* = (r_H \lambda)^{-1}, \tag{1}$$

and RH-268 proves $a_n / q_*^n \to 1$ and hence the sharp target radius [9, 13]. None of these is a moving noisy coefficient theorem. The noisy all-order bridge and the canonical physical determinant required by Gate A remain open.

For the present batch, freeze

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \qquad H_k = k R^{-2k}, \qquad R = \frac{7}{5}, \qquad u = 4k. \tag{2}$$

The upper endpoint is admissible for the one-alias convention $2k < u \le 4k$: the order $4k$ is outside the strict prefix $n < u$. All comparisons below use the Hardy full-trace constituent and the corrected folded basepoint observation of RH-334. No forward probability law is silently substituted for a cyclic trace.

## 2 Ten typed layers

The first nine sources are [1–3, 5–8, 10, 12]. Their scopes are retained literally.

| paper | strict result | boundary retained in this review |
|---|---|---|
| RH-332 | Sharp positive order-$\sigma$ coefficient for the second physical hybrid row, separately in both repelling orientations. | Local row theorem; not a fully physical versus fully affine two-leg equality, all-cycle transport, or cyclic trace estimate. |
| RH-333 | Positive retained-preclosing-path $L^1$ escape gap for the raw mass-one full-line forward affine tube. | Stops only that reference and retained coordinate; cyclic bridges, adapted references, and the endpoint after the omitted closing row remain untested. |
| RH-334 | Corrected signed/folded basepoint localization and the exact Hardy full-trace observation identity. | Individual windows are frozen gauges; the full-trace constituent is not the modulus complement without the head defect $d$. |
| RH-335 | Exact projector-localized parity ledger and cellwise extension nonuniqueness. | The projector allocation is a signed gauge, not a physical probability density; the required physical adapted-norm upper exponent is not supplied. |
| RH-336 | Exact projector-mass threshold and a positive row-stochastic isospectral cell-motion family. | The family is finite nonphysical algebra; it does not decide physical $\Delta_B + \Delta_S$ cancellation. |
| RH-337 | Exact proof that RH-329 uses a faster-than-physical clock and hence has phase tending to $-\infty$. | The hatted comparator is rejected only for bounded-phase physical replacement; correct-clock target accuracy remains not testable. |
| RH-338 | A physical critical far-orbit atom of alias size and a necessary signed diffuse compensation law. | The atom is not an aggregate far lower bound because the signed diffuse remainder may compensate it. |
| RH-339 | Every one-alias cut contains the first lower sideband $2k-2$ and its physical orbit atom. | Off-alias vanishing requires signed compensation; no lower bound for the fully signed sideband coefficient is obtained. |
| RH-340 | The two analytic tails close at $u = 4k$, and the direct/full-trace prefix synchronization inequality is exact. | Prefix, critical, off-alias, and head budgets remain open, so RH-288 is inactive. |
| RH-341 | Ten-layer source audit and abstract signed-completion underdetermination theorem. | Information-class theorem only; no pair of physical noisy operators is constructed. |

The batch therefore contains ten rigorous scoped conclusions and zero discharged aggregate actual-replacement obligations. In particular, RH-332 is not promoted to a two-leg/all-cycle theorem, RH-334 is not promoted to modulus-complement identification, and the nonphysical RH-335–RH-336 fixtures are not promoted to noisy-operator counterexamples.

## 3 Exact synchronized prefix reduction

For every $n \geq 2$, the corrected all-order constituent is

$$q_{\sigma,k,n} = \mathcal{B}_{\sigma,k,n} + \mathcal{S}_{\sigma,k,n} + \mathcal{R}_{\sigma,k,n} + \mathcal{P}_{\sigma,n} - \mathcal{A}_{k,n}. \tag{3}$$

RH-339 states this all-order proposition explicitly; RH-334 supplies the underlying all-order raw partition and the first-alias typed observation. Let

$$p_{\sigma,k,n} := \tau_{\sigma,n} - a_n = q_{\sigma,k,n} - d_{\sigma,k,n}, \qquad d_{\sigma,k,n} = h_{\sigma,n} - s_{k,n}. \tag{4}$$

At a common finite cut $u$, define

$$P_u(R) = \sum_{2\le n<u} \frac{|p_{\sigma,k,n}|R^n}{n}, \qquad E_u(R) = \sum_{2\le n<u} \frac{|q_{\sigma,k,n}|R^n}{n}, \qquad D_u(R) = \sum_{2\le n<u} \frac{|d_{\sigma,k,n}|R^n}{n}. \tag{5}$$

**Proposition 3.1** (same-clock prefix synchronization)**.** *For every common finite cut and radius,*

$$|P_u(R) - E_u(R)| \le D_u(R). \tag{6}$$

*Consequently $D_u(R) \to 0$ implies $P_u(R) \to 0$ if and only if $E_u(R) \to 0$.*

*Proof.* Equation (4) and the reverse triangle inequality give $\big||p_n| - |q_n|\big| \le |d_n|$ at each order. Multiply by $R^n/n$ and sum. This is the exact RH-340 argument. □

At $u = 4k$, RH-340 re-applies the RH-282 mass estimate and the deterministic envelope (1) to prove that both analytic tails vanish. The one-alias extraction is

$$E_{4k}(R) = E_{\mathrm{off},4k}(R) + \frac{|q_{\sigma,k,2k}|}{2H_k}. \tag{7}$$

Hence the remaining sufficient input for the RH-288 prefix leaf is exactly

$$D_{4k}(R) \to 0, \qquad E_{\mathrm{off},4k}(R) \to 0, \qquad q_{\sigma,k,2k} = o(H_k). \tag{8}$$

No term in (8) is proved by the current batch. Tail closure alone therefore does not activate determinant gluing.

# 4 Two mandatory signed compensations

Let $m = k - 1$. RH-338 and RH-339 prove physical signed orbit constituents

$$\mathcal{R}_{k,\mathrm{orb}} = -D_k^{\mathrm{orb}}, \qquad \frac{D_k^{\mathrm{orb}}}{H_k} = \frac{2}{C_M}(\beta R)^{2k}\{1 + o(1)\} \to +\infty, \tag{9}$$

$$\mathcal{R}_{k,-,\mathrm{orb}} = -D_m^{\mathrm{orb}}, \qquad \frac{D_m^{\mathrm{orb}}}{H_m} = \frac{2}{C_M}(\beta R)^{2m}\{1 + o(1)\} \to +\infty. \tag{10}$$

Here $H_m = mR^{-2m}$ and $\beta R > 1$ has an exact repository certificate. Define the signed complements inside the Hardy full-trace coefficients by

$$C_k^0 = q_{\sigma,k,2k} + D_k^{\mathrm{orb}}, \qquad C_k^- = q_{\sigma,k,2m} + D_m^{\mathrm{orb}}. \tag{11}$$

Using (4), the two direct coefficients are exactly

$$p_{\sigma,k,2k} = -D_k^{\mathrm{orb}} + C_k^0 - d_{\sigma,k,2k}, \tag{12}$$

$$p_{\sigma,k,2m} = -D_m^{\mathrm{orb}} + C_k^- - d_{\sigma,k,2m}. \tag{13}$$

**Theorem 4.1** (necessary adjacent-order compensation)**.** *If $P_{4k}(R) \to 0$, then*

$$C_k^0 - d_{\sigma,k,2k} = D_k^{\mathrm{orb}} + o(H_k), \tag{14}$$

$$C_k^- - d_{\sigma,k,2k-2} = D_{k-1}^{\mathrm{orb}} + o(H_{k-1}). \tag{15}$$

*The relative precisions are respectively $o((\beta R)^{-2k})$ and $o((\beta R)^{-2(k-1)})$. If also $D_{4k}(R) \to 0$, then the two displayed head terms are negligible on their target scales.*

*Proof.* Both orders occur in the nonnegative prefix, and

$$P_{4k}(R) \geq \frac{|p_{\sigma,k,2k}|}{2H_k} + \frac{|p_{\sigma,k,2m}|}{2H_m}.$$

Prefix closure makes the two numerators $o(H_k)$ and $o(H_m)$. Substitute (12). Division by the two asymptotics in (9) gives the relative precisions. The same nonnegative-summand argument applied to $D_{4k}$ makes the two head terms negligible. □

Taking separate absolute values before the signed sums necessarily contains

$$\frac{D_k^{\mathrm{orb}}}{2H_k} + \frac{D_m^{\mathrm{orb}}}{2H_m} = \frac{1}{C_M}\left((\beta R)^{2k} + (\beta R)^{2k-2}\right)\{1 + o(1)\} \longrightarrow +\infty. \tag{16}$$

Thus the cancellation-blind orbit/diffuse/head route is stopped. Equation (16) is not a lower bound for the fully signed prefix.

# 5 Abstract signed-completion underdetermination

Put

$$X_k^0 = C_k^0 - d_{\sigma,k,2k}, \qquad X_k^- = C_k^- - d_{\sigma,k,2k-2}. \tag{17}$$

The current source information fixes the two atoms and the exact identities

$$p_{\sigma,k,2k} = -D_k^{\mathrm{orb}} + X_k^0, \qquad p_{\sigma,k,2k-2} = -D_{k-1}^{\mathrm{orb}} + X_k^-, \tag{18}$$

but supplies no moving-order estimate for either signed combined complement. Define the *source information class* to consist of abstract signed coefficient ledgers satisfying the proved identities, scales, and data-type labels, with no additional physical realizability assumption.

**Theorem 5.1** (opposite abstract completions)**.** *Within the source information class, neither $P_{4k}(R) \to 0$ nor $P_{4k}(R) \not\to 0$ is logically determined. More sharply, there is an abstract completion with zero direct prefix and another whose two mandatory orders alone diverge in the weighted prefix. Both may be chosen with $d_{\sigma,k,n} = 0$ at every order.*

*Proof.* For the cancelling completion, set

$$X_k^0 = D_k^{\mathrm{orb}}, \qquad X_k^- = D_{k-1}^{\mathrm{orb}},$$

and set every other direct prefix coefficient to zero. Equations (18) give $P_{4k} = 0$.

For the noncancelling completion, set $X_k^0 = X_k^- = 0$ and again set every other direct prefix coefficient to zero. Then

$$P_{4k}(R) \geq \frac{D_k^{\mathrm{orb}}}{2H_k} + \frac{D_{k-1}^{\mathrm{orb}}}{2H_{k-1}} \longrightarrow +\infty$$

by (16). Taking $d = 0$ realizes both choices as abstract full-trace complement ledgers. The current source class imposes no theorem that excludes either signed choice, so neither aggregate alternative is a logical consequence of the available information. □

*Remark* 5.2 (strict interpretation)*.* The proof constructs no Markov kernel, transfer operator, noisy quadratic map, or determinant. It does not assert that either completion is physically realizable. Its content is exactly the insufficiency of the currently proved signed information. A physical verdict requires a new moving-order theorem for the actual combined complements and head defect.

**Corollary 5.3** (frontier verdict)**.** *The batch proves a scoped failure of separate-absolute splitting and an abstract information-class underdetermination theorem. Aggregate physical prefix closure, aggregate nonclosure, $E_{\mathrm{off},4k}$, the head budget, and RH-288 activation remain `NOT_TESTABLE` or open.*

## 6 Route coordinate, archive, and claim boundary

The unique repository-backed coordinate after this audit is

`synchronized_actual_first_alias_signed_completion_open`.

The narrow next theorem must estimate the actual $X_k^0$, $X_k^-$, the rest of the off-alias prefix, and the head defect on the same clock and Hardy data type. A finite fit, the rejected RH-329 clock, a nonphysical similarity family, a forward retained path without cyclic identification, or an abstract completion is not a reopening input.

The executable review records ten scoped conclusions, zero discharged aggregate replacement obligations, four exact algebraic completion witnesses, and seventeen false forbidden claims. Finite rows are formula checks only. The individual archive contains fifteen publication files for each of RH-332–RH-340 and nineteen for RH-341. The batch manifest therefore contains $9 \cdot 15 + 19 = 154$ publication files; after per-paper and batch metadata, the ten controlled paper trees contain 176 files. Both individual and batch verifiers report zero failures.

The nine upstream result ledgers contain forty-five false Gate values. This review adds five more, so all fifty Gate A–E values remain false. Gate A still lacks a canonical intrinsic dynamical spectral determinant; Gates B–E remain independent open obligations. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

## References


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